

Jinhee Lee

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RESEARCH INTERESTS

Computer Vision, Vision-Language Models, Robotics, Foundation Models, Generative AI, and Bioinformatics

Focus Areas: Surgical AI, Medical Imaging, Clinical Vision Question Answering, Human-Robot Interaction, Autonomous Robotic Surgery, Bio-inspired Robotics, and Healthcare

EDUCATION

Ohio State University | Columbus, OH, USA

M.S. in Computer Science and Engineering

2019 - 2021

Courses: Machine Learning, Neural Network, NLP, Advanced Artificial Intelligence, Graphics, Algorithms, and OS

University of Science and Technology | Korea Aerospace Research Institute | Daejeon, Korea

M.Eng. in Aerospace System Engineering

2016 - 2018

Thesis: Study on the Spray Characteristics of Fuel Injectors For Scramjet Engines

Courses: Sensor Signal Processing

Inha University | Incheon, Korea

B.S. in Aerospace Engineering

2010 - 2016

Project: Rocket-Launched Unmanned Aerial Vehicle (UAV) Design and Flight Test of a Scaled Prototype

Courses: Engineering Mechanics, Control Systems, Electromagnetics, Chemistry, Linear Algebra, and Vector Calculus

*Honors: Junior Year Top Student (Full Scholarship) **Military: 2 years of service, Republic of Korea Air Force

PROFESSIONAL EXPERIENCE

AI Research Assistant | Stanford University

2024 - Present

Surgical AI

- Developed an intraoperative brain surgery segmentation model, improving arachnoid membrane IoU by 7.35%
- Performed intraoperative absolute depth estimation for MVD surgery, achieving <2 mm error
- Co-first-authored two surgical AI research papers, including impactful medical AI conference publications
- Exploring Surgical Vision-Language Foundation Models and AI Preference Alignment

AI Research Collaborator | Bongseng Memorial Hospital

2024 - Present

Medical Data Curation and AI Research

- Conducted AI research on brain CT/MRA/MRI for neurovascular classification and brain-lesion segmentation
- Supported neurosurgical imaging data curation and set up an AI computing server

AI Engineer | Lululab Inc. | Seoul, Korea

2022 - Present

Segmentation, Detection, and Classification Model Development

- Enhanced facial acne segmentation with GAN-generated synthetic patches, reaching a dice score of 71.47%
- Developed YOLO-based OCR for blood pressure registration for Digital Therapeutics, achieving 94.7% accuracy
- Implemented AI-driven personal color analysis model with 85.8% accuracy and created an demo app

Generative AI

- Developed facial aging and skin disease progression modules using Stable Diffusion model and GAN-based model
- Implemented a face anonymization system leveraging a diffusion model for privacy-preserving medical data
- Developed a ChatGPT-based skin chatbot with GPT fine-tuning, function calling, and RAG

Recommendation Systems

- Designed recommendation systems for cosmetics, skin treatments, and life routines
- Developed a deep learning model for cosmetic ingredients analysis

Aircraft Engine Researcher | Korea Aerospace Research Institute | Daejeon, Korea

2016 - 2018

- Supersonic Combustion and Turbojet Engine Research

Research Intern | Korea Aerospace Research Institute | Daejeon, Korea

2014 Summer

- Aircraft engine compressor research

Aircraft Mechanic | Republic of Korea Air Force | Busan, Korea

2011 - 2013

- Repair and maintenance of the E-737 AEW&C aircraft

PUBLICATIONS AND PATENT

[1] Jinhee Lee, Hwanhee Lee, Jay J. Park, Jeong Woo Ahn, Jong Yun Kwon, Ciara McMahon, Julia Lewandowski, Seohee Park, Sanghoon Lee, and Vivek P. Buch. *Arachnoid Membrane Segmentation in Intraoperative Microscopic MVD Surgery Scenes*. In *MICCAI WorkShop (COLAS)*. Springer Nature Switzerland, 2025. (Accepted and in press)

[2] Jinhee Lee, Chanhyuk Lee, Huisu Yoon, Semin Kim, and Jongha Lee. *Realistic Skin Trouble Simulation via Image Generation Models*. In *IEEE International Conference On Image Processing (ICIP)*. IEEE, 2025. [Oral Presentation]

[3] Jinhee Lee, Hwanhee Lee, Jay J. Park, Ethan Htun, Bruce Changlong Xu, Sanghoon Cho, Sanghoon Lee, and Vivek P. Buch. *Intraoperative Absolute Depth Estimation in MVD Surgery*. In *IEEE 38th International Symposium on Computer-Based Medical Systems (CBMS)* (pp. 341-342). IEEE Computer Society, 2025.

[4] Semin Kim, Jinhee Lee, Chanhyuk Lee, and Jongha Lee. *Improving Facial Acne Segmentation through Semi-Supervised Learning with Synthetic Images*. *Journal of Korea Multimedia Society*, 27(2), 241-249, 10.9717/kmms.2024.27.2.241

[5] Jinhee Lee, Huisu Yoon, Semin Kim, Chanhyuk Lee, Jongha Lee, and Sangwook Yoo. *Deep learning-based skin care product recommendation: A focus on cosmetic ingredient analysis and facial skin conditions*. *Journal of Cosmetic Dermatology*, 23(6), 2066-2077.

[6] Jinhee Lee, Sanghoon Lee, Kyungjae Lee, Jaiho Kim, and Soo-Seok Yang. *Study on Spray Visualization and Atomization Characteristics of Air-assist Type Injector for Scramjet Engine*. *Journal of the Korean Society of Propulsion Engineers*, 21(5), 88-96.

[P1] Method and Apparatus for Recommending Cosmetics Suitable for Users Based on the Efficacy of the User's Skin Type and Cosmetic Ingredients, Korean: 10-2022-0079131/10-2450424, PCT: PCT/KR2023/008926

TEACHING AND MENTORING EXPERIENCE

Mentoring | Lululab Inc. - Sungkyunkwan University

2022 - 2024

Led and mentored 9 undergraduate students across 3 AI internship projects

- OCR for cosmetic ingredients recognition
- ChatGPT-based skincare chatbot system development
- Personal color analysis and makeup product recommendation

TECHNICAL SKILLS

Programming Languages: Python (Advanced), Java (Intermediate), C++ (Intermediate), MATLAB (Intermediate), C#

Software and Tools: Pytorch, Opencv, Git, Linux, Docker, Unity Engine, Android Studio, Blender, 3ds Max, LaTeX